ABHISHEK DILEEP

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Summary

Audio machine learning engineer with two years at Sony India's Audio Business Unit, where spatial-audio and source-separation work shipped in product releases deployed worldwide. Currently an M.Tech (Intelligent Systems) researcher at IIT Mandi working on acoustic scene classification, unsupervised domain adaptation, and end-to-end speech-to-speech models, with a first-author IEEE publication.

Education

IIT, Mandi Mtech AI — *CGPA: 8.00 | 2024 – Present*
NMAMIT, Nitte B.Tech CSE — *CGPA: 8.76 | 2017 – 2021*
DIAT, Pune Certificate ML Engineering — *CGPA: 8.11 | Nov 2023*

Publications

- **Device Invariance for Acoustic Scene Classification using Domain Adaptation**, *IEEE Conference on Engineering Informatics (ICEI 2026)*, IIT Dharwad, India, Nov 2026. [arXiv:2607.25887]

Professional Experience

DRDO | Project Associate — 2025 – Present

- Building an end-to-end **sonar classification system** for naval sonar data with model design, training.
- Used **Bellhop** underwater acoustic propagation model for underwater depth modelling, enabling domain-robust classification across deployment condition with **95% overall accuracy**

Sony SARD India | ML Engineer — 2021 – 2023

- Developed **Ear Detection model** for spatial-audio signal-processing modules for **Sound Personalisation**, shipped in a worldwide product release; quantized model to **20%** from 10Mb to 2Mb of original size.
- Built multiple POCs for **music source separation**, Sony graphical sound suite.

Research Experience

Manas LAB, IIT Mandi | Research Scholar — 2024 – 2025

- Built **unsupervised domain adaptation (UDA)** pipelines for cross-device **acoustic scene classification** on the DCASE 2020 Mobile Urban Acoustic Scenes benchmark, with a ∼**10% accuracy gain** over the baseline.

Projects

- **LLaMA-Omni2.5 — End-to-End Speech-to-Speech Model**
  Coupled Whisper encoder to Qwen2.5 LLM for speech-in speech-out dialogue.
- **Domain Adaptation for Acoustic Scene Classification**
  Compared five adaptation methods achieving device-invariant acoustic scene classification.
- **Low-Dose to High-Dose CT Image Synthesis**
  Used GANs to synthesise high-dose CT scans from low-dose inputs.
- **Quantised Side-Tuning for Parameter-Efficient LLM Fine-Tuning**
  Reduced trainable parameters for LLaMA-7B while preserving task performance.

References

- **Prof. Padmanabhan Rajan** — **Professor** | IIT Mandi